

Pathway Analysis of RNA-seq Data: Reducing Redundancy and False Positives

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MaRBLLe 2.0 Semester 2 Report

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June 2026

Abstract This report evaluates and extends a pathway analysis framework for RNA-seq enrichment results. The main goal was to test whether two pipelines developed in the first semester could generalise beyond the original dataset and still produce compact, interpretable pathway shortlists. The pipelines were applied to several RNA-seq datasets with different levels of differential expression, pathway coverage, and overlap between Gene Ontology and WikiPathways. The results showed that the framework could be applied across datasets, but that the reliability of the final outputs depended strongly on dataset properties. A dataset-aware threshold policy improved Pipeline A in borderline cases by relaxing the consensus threshold while keeping the stability requirement strict, but it also showed that low coverage or missing cross-collection overlap cannot be solved only by changing thresholds. Pipeline B was evaluated with an uncapped ablation study, which showed that overlap clustering had the strongest effect on the candidate pool, while the final top-N cap mainly acted as a presentation choice. Comparing both pipelines showed that they are complementary rather than redundant: Pipeline A is strongest when a reliable DE gene list is available, while Pipeline B adds directionality and can be useful when gene-level differential expression is limited. Finally, a dashboard was developed to make the final pathway results easier to inspect, compare, and interpret.

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1 Introduction

RNA sequencing (RNA-seq) is widely used to study changes in gene expression between biological conditions. A standard RNA-seq analysis often produces long lists of differentially expressed genes, together with statistics such as fold changes and adjusted p-values. Although these gene-level results are useful, they are difficult to interpret directly, especially when the goal is to understand which biological processes or disease mechanisms are involved.

Pathway and Gene Ontology (GO) enrichment analysis are commonly used to make RNA-seq results more interpretable by summarising gene-level changes into biological processes, pathways, or functional terms [5, 2]. WikiPathways (WP) provides an additional pathway collection that can be used together with GO to interpret gene expression results [1]. However, enrichment analysis can also produce long and redundant result

lists. Many pathways and GO terms share genes or describe closely related biological processes, which can make the final output difficult to interpret. In addition, some enrichment methods do not fully account for gene–gene correlation, which can increase the risk of false positive pathway findings [11].

In the first semester of this project, two pathway analysis pipelines were developed to reduce redundancy and improve interpretability [3]. Pipeline A was based on over-representation analysis, semantic redundancy reduction, overlap clustering, bootstrap stability, and cross-collection consensus. Pipeline B used correlation-aware and ranked gene-set methods, mainly CAMERA and fgsea, followed by agreement filtering and redundancy reduction. These methods build on existing approaches for gene-set analysis, semantic similarity, and correlation-aware pathway testing [5, 9, 6, 10, 11]. This first version showed that structured filtering can reduce large enrichment outputs into more compact pathway shortlists.

The aim of the second semester was to test whether this approach generalises beyond the original dataset. The pipelines were applied to multiple RNA-seq datasets with different characteristics, including datasets with strong signal, weak signal, low pathway coverage, and limited overlap between gene-set collections. This made it possible to study when the pipelines produce reliable results and when the final shortlist is limited by the input data.

This semester also focused on improving the framework around the pipelines. A dataset-aware threshold policy was developed for Pipeline A, supported by thresh-

old sweeps and simulation experiments. Pipeline B was investigated through an ablation study to understand which components most strongly affect the final shortlist. Finally, the outputs were integrated into a Shiny dashboard to make the results easier to inspect, compare, and interpret.

Overall, this report focuses on evaluating and extending the pathway analysis framework developed in the first semester. The main goal is to understand how RNA-seq pathway enrichment results can be made more reliable, less redundant, and easier to interpret across different datasets.

1.1 Research Questions

This semester, the project was guided by the following research questions:

1. How well do the two pathway analysis pipelines generalise across RNA-seq datasets with different levels of differential expression, pathway coverage, and gene set overlap?
2. Can dataset-specific threshold selection improve the reliability of Pipeline A compared to using fixed stability and consensus thresholds?
3. Which components of Pipeline B contribute most to producing a compact and biologically interpretable pathway shortlist?
4. Are Pipeline A and Pipeline B complementary across different dataset types, or does one pipeline consistently produce more useful pathway results?
5. Can an interactive dashboard make the final pathway results easier to inspect, compare, and interpret?

2 Previous Work

The first semester of this project established the methodological foundation for the current work. The aim was to develop pathway analysis pipelines that could reduce redundancy and false positives in RNA-seq enrichment results, while still preserving biologically meaningful signal. Two pipelines were implemented and compared. Although they had the same general goal, they used different enrichment strategies and therefore represented different approaches to pathway interpretation.

Pipeline A was designed as a trust-first pipeline based on over-representation analysis (ORA). It started from a list of differentially expressed genes and tested whether these genes were overrepresented in GO Biological Process terms and WP pathways [2, 1]. ORA is a commonly used and interpretable enrichment approach, usually based on Fisher’s exact or hypergeometric testing [5]. However, it depends on the definition of a differentially expressed gene list, meaning that the final pathway output can be affected by the chosen gene-level

cutoff. To improve the reliability of the output, Pipeline A combined ORA with several additional filtering steps: semantic similarity reduction, overlap-based clustering, bootstrap stability analysis, and cross-collection consensus.

Pipeline B was developed as a more conservative and correlation-aware alternative. Instead of relying only on a binary differentially expressed gene list, it used ranked gene-level information through CAMERA and fgsea. CAMERA was included because it accounts for inter-gene correlation within gene sets, which is important because genes in the same pathway are often co-expressed and therefore not independent [11]. fgsea was used as a GSEA-style ranked enrichment method that can detect coordinated pathway-level changes without requiring a hard differential expression cutoff [9, 6]. In Pipeline B, pathways were prioritised when they were supported by both methods and showed consistent enrichment direction.

A central part of both pipelines was redundancy reduction. Enrichment results often contain many overlapping terms, especially when GO terms describe related biological processes at different levels of specificity [7, 8]. In Pipeline A and Pipeline B, this was addressed in two ways. First, semantic similarity was used to reduce redundancy among GO terms based on their position in the ontology structure [10]. Second, overlap-based clustering was used to reduce redundancy between gene sets that shared many of the same genes. These two approaches target different forms of redundancy: semantic similarity is based on biological meaning within the ontology, while overlap clustering is based on shared gene membership.

The first semester results showed that structured filtering could reduce large enrichment outputs into more compact and interpretable pathway shortlists. Pipeline A retained a broader biological signal and was useful for exploratory analysis, while Pipeline B produced a smaller and more focused output. Therefore, the pipelines were interpreted as complementary strategies rather than as direct replacements for each other.

However, the first-semester work was mainly a proof of concept on one dataset. Several questions remained open. It was not clear whether the same pipeline behaviour would generalise to other RNA-seq datasets, whether fixed threshold choices were appropriate across different data conditions, or which Pipeline B components were most important for the final shortlist. These limitations motivated the second-semester work, which shifted the focus from pipeline construction to multi-dataset evaluation, dataset-aware threshold selection, component testing, cross-pipeline comparison, and dashboard development.

3 Data

This semester, the pipelines were evaluated on several RNA-seq datasets rather than only on the original dataset from the first semester. The aim was to test whether the framework could be applied across datasets

that differed in biological comparison, gene identifier type, transcriptome coverage, and differential expression signal.

The first dataset, referred to as `dataset_0`, was the original dataset from the first semester. It was kept as the reference dataset because the pipelines had already been developed and initially tested on it. This made it useful as a baseline for comparing the extended second-semester analyses.

Three additional datasets, referred to as `dataset_1`, `dataset_2`, and `dataset_3`, were used as a blinded study. These datasets were included to test whether the pipelines could handle new input data without being tuned only to the original dataset. They also introduced more variation in the strength of the differential expression signal and in the number of genes that could be linked to pathway annotations.

In addition, several datasets were provided through the collaboration with the Biochemistry department: `ACTA2K0_iVSMC`, `Aneurysm_pVSMC`, `PAR1K0_iVSMC`, and `Lineages_iVSMC`. These datasets were especially useful because they came from the biological research context of the project and used ENSEMBL gene identifiers, which required conversion before pathway analysis. The `Lineages_iVSMC` dataset was inspected, but excluded from the final analysis because it showed almost no meaningful differential expression signal. Running enrichment analysis on this dataset would therefore have produced little interpretable pathway-level information.

For each analysed dataset, the available inputs included normalised gene expression data, sample metadata, and gene-level differential expression statistics. Pipeline A used the differential expression statistics to define the input gene list for over-representation analysis. Pipeline B used the expression matrix and sample metadata to perform ranked and correlation-aware gene-set testing. Because the datasets used different gene identifier types, genes were converted to a consistent identifier format where needed before matching them to pathway annotations.

A dataset-specific enrichment universe was defined before running the pathway analysis. In this project, the universe was the intersection of genes measured in the RNA-seq data and genes present in the pathway annotation databases. This was necessary because enrichment should only be tested against genes that could actually have been observed in the experiment. GO Biological Process terms and WP pathways were used as the pathway annotation sources throughout the analysis [2, 1].

4 Methods

The methods in this semester were organised around evaluating and extending the pipeline framework developed in the first semester. The goal was not to redesign the two pipelines from scratch, but to test how they behave across different datasets, automate part of the threshold selection process, and analyse which methodological choices most strongly affect the final pathway

shortlist. The main methodological additions were the multi-dataset evaluation, the dataset-aware threshold policy, simulation experiments, a Pipeline B ablation study, a cross-pipeline comparison, and the development of an interactive dashboard.

4.1 Evaluation across datasets

To assess whether the pipeline framework generalised beyond the original dataset, Pipeline A and Pipeline B were applied to all analysed RNA-seq datasets. The same core pipeline logic was kept fixed as much as possible, so that differences in output could be interpreted as dataset-dependent behaviour rather than as changes in the method itself.

For each dataset, the analysis recorded the dataset-specific universe, the number and fraction of differentially expressed genes, and the final output type produced by each pipeline. For Pipeline A, outputs were grouped into three confidence levels. Tier 1 pathways were stable pathways that also formed GO-WP cross-collection consensus pairs. Tier 2 pathways were stable pathways without sufficient cross-collection consensus. Candidate pathways were lower-confidence outputs used when the data did not support a stronger shortlist. For Pipeline B, the recorded outputs included the agreement mode, direction of enrichment, and final selection reason. These outputs were compared using the same summary criteria across datasets, allowing the behaviour of the pipeline framework to be evaluated consistently.

4.2 Dataset-aware threshold policy

One goal of the second semester was to make the final selection in Pipeline A less dependent on one fixed threshold setting. In the first semester, the same stability and consensus thresholds were applied to the original dataset. This worked as a proof of concept, but it was not clear whether the same settings would also be appropriate for datasets with different levels of differential expression, pathway coverage, or overlap between GO and WP. Therefore, a dataset-aware threshold policy was developed for the final stages of Pipeline A.

The policy focused on two thresholds. The first was the bootstrap stability threshold, denoted as τ , which defines how often a pathway must be selected across bootstrap iterations to be considered stable. The second was the consensus Jaccard threshold, denoted as `CONS_J`, which defines the minimum gene-set overlap required between a stable GO term and a WP term for cross-collection consensus. Gene-set overlap and semantic similarity are commonly used to identify redundancy between enriched terms [7, 8, 10]. These thresholds were chosen because they directly affect whether the final Pipeline A output is classified as Tier 1, Tier 2, or candidates.

4.2.1 Threshold sweep

To test threshold sensitivity, a small grid search was applied to each dataset. The stability threshold was tested at $\tau \in 0.50, 0.60, 0.70$, and the consensus Jaccard threshold was tested at $\text{CONS_J} \in 0.10, 0.15, 0.20$. This gave nine threshold combinations per dataset.

The sweep was applied only to the downstream stability and consensus logic. The earlier steps of Pipeline A, including over-representation analysis, semantic collapse, overlap clustering, and the original bootstrap runs, were not rerun for every threshold combination. Instead, the existing representative pathways and bootstrap evidence were reused. For each threshold combination, the number of Tier 1 pathways, Tier 2 pathways, and GO-WP consensus pairs was recorded.

4.2.2 Dataset classification and threshold selection

Based on the sweep results and dataset-level properties, each dataset was assigned to one of four practical classes. The classes were applied in priority order. A dataset was classified as *low coverage* if the universe size was below 2000 genes or if the median fraction of pathway genes measured was below 0.20. A dataset was classified as *no overlap* if no GO-WP consensus pairs were found in any threshold combination and Tier 1 was never achievable. A dataset was classified as *borderline* if the default setting did not produce a sufficient Tier 1 shortlist, but Tier 1 became achievable at another threshold combination. Finally, a dataset was classified as *healthy* if the default thresholds already produced at least five Tier 1 pathways.

The final thresholds were then selected according to the dataset class. For healthy datasets, the default setting was kept, with $\tau = 0.70$ and $\text{CONS_J} = 0.15$. For borderline datasets, the stability threshold was kept at $\tau = 0.70$, while CONS_J was relaxed to the smallest value that produced a sufficient Tier 1 shortlist, usually 0.10. For no-overlap datasets, τ was also kept at 0.70, because the issue was not pathway instability but missing overlap between GO and WP. For low-coverage datasets, a more permissive setting was allowed, but the output was flagged as lower confidence because limited pathway coverage cannot be fully corrected by threshold adjustment.

Overall, the policy separated datasets where threshold relaxation could reasonably recover a high-confidence shortlist from datasets where the input data itself limited the reliability of the enrichment result. In this way, the policy acted both as a threshold selection method and as a diagnostic tool for interpreting Pipeline A outputs.

4.3 Simulation experiments for threshold behaviour

Simulation experiments were used as a small methodological check for the threshold policy. They did not use

real RNA-seq expression values and were not meant to validate specific biological pathways. Instead, they used artificial gene universes, artificial pathways, and simulated DE gene sets to test how simple dataset properties affect bootstrap stability in Pipeline A.

Two simulation settings were used. The first tested pathway coverage by changing how much of an artificial pathway was actually observed in the dataset. This was meant to represent a case where only part of a pathway is measured, for example in a focused panel. The second tested pathway signal strength by changing both the total size of the DE gene set and the number of DE genes inside the pathway. This made it possible to create weaker and stronger pathway signals in a controlled way.

For each setting, ORA was repeated under bootstrap sampling. Stability was then calculated as the fraction of bootstrap iterations in which the artificial pathway remained significant. These simulations were used to check whether low pathway coverage or weak pathway-level signal could explain unstable enrichment results, without adding the extra complexity of real biological data. The simulation results are shown in Supplementary Figure 7.

4.4 Pipeline B ablation study

An ablation study was performed to investigate which components of Pipeline B most strongly affected the final candidate pool. This was done by removing or bypassing one component at a time and comparing the resulting pathway sets to the full Pipeline B output. The ablation was performed on `dataset_0`, because this was the original reference dataset and provided a stable baseline where Pipeline B produced a full directional shortlist. This made it suitable for testing how much the output changed when individual components were removed.

The main ablation analysis focused on uncapped Pipeline B outputs. In the standard Pipeline B workflow, the final output is limited by a top-N cap per direction. This is useful for producing a compact shortlist, but it can hide differences between pipeline variants. Therefore, the uncapped ablation compared the full set of pathways that passed the final selection criteria before applying the presentation cap.

Four variants were compared. The first was the full uncapped Pipeline B, where all components were active. The second removed GO semantic collapse, so GO terms passed to the next step without ontology-based redundancy reduction. The third removed Jaccard overlap clustering, so pathways were not grouped into overlap-based representatives before final selection. The fourth was a diagnostic CAMERA-only selection variant, where fgsea agreement was bypassed to test how much of the candidate pool could be recovered from CAMERA alone. This diagnostic variant was not treated as a replacement for the full pipeline, but as a way to test the contribution of the CAMERA-fgsea agreement step.

The ablation variants were defined by removing or bypassing individual steps from the Pipeline B workflow shown in Figure 1.

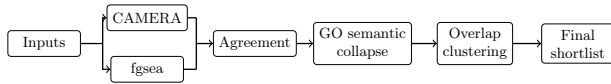


Figure 1: Pipeline B workflow.

The variants were compared using several summary measures: the number of pathways retained in the final uncapped pool, the UP/DOWN balance, the GO/WP composition, and pathway ID overlap with the full uncapped pipeline. Pairwise Jaccard similarity was used to compare the final pathway sets between variants. Because Jaccard similarity can be affected by different pool sizes, containment measures were also considered, such as the percentage of full-pipeline pathways retained in each variant and the percentage of variant pathways also present in the full pipeline.

4.5 Pipeline A vs Pipeline B comparison

Pipeline A and Pipeline B were compared to investigate whether they produced overlapping or complementary pathway results. This comparison was important because the two pipelines are based on different assumptions. Pipeline A is ORA-based and uses a DE gene list, bootstrap stability, and GO-WP consensus to produce stable but undirected pathway shortlists. Pipeline B uses CAMERA and fgsea-style ranked enrichment on gene-level information, does not require a DE gene pre-filter, and produces direction-specific UP and DOWN pathway shortlists [11, 9, 6].

The comparison was performed on selected datasets where both pipelines produced interpretable outputs. For each dataset, the final Pipeline A shortlist was compared with the final Pipeline B shortlist. First, exact pathway ID overlap was calculated. This was a strict comparison, because it only counted cases where both pipelines selected the same GO term or WP pathway. However, exact overlap was not expected to be high, since the pipelines use different statistical methods, different filtering logic, and different output structures.

To avoid interpreting exact ID overlap as the only possible form of agreement, the comparison also included a descriptive check of broader thematic similarity. This was not treated as a formal statistical test. Instead, pathway names, gene-set collections, and general biological descriptions were inspected to identify cases where the two pipelines pointed to related pathway themes even if the exact pathway IDs differed. For shared pathway IDs, Pipeline A stability scores were compared with Pipeline B direction and enrichment strength. This made it possible to identify pathways that were supported by both pipelines and to assess whether Pipeline B added directional information to findings that Pipeline A detected as stable.

The comparison was also used to describe situations where one pipeline was more informative than the other.

Because Pipeline A and Pipeline B use different inputs and statistical assumptions, their outputs were interpreted in relation to dataset properties such as DE gene fraction, pathway coverage, and the number of samples available for Pipeline B. Therefore, the comparison did not aim to choose one universally better pipeline, but to determine under which data conditions each pipeline provided more useful information.

4.6 Dashboard development

An interactive dashboard was developed in Shiny as an inspection layer for the final pathway shortlists [4]. It was designed to load precomputed outputs from Pipeline A, Pipeline B, the threshold policy, and the cross-pipeline comparison analysis, rather than rerunning the pipelines. This made it possible to explore the results in one place instead of manually comparing multiple output files.

The dashboard allows the user to select a dataset and pipeline, inspect the final shortlist in a table, view summary information, and compare results using visualisations such as a dot plot, selection funnel, overlap network, and A vs B comparison view. It also includes a detail panel for selected pathways and an option to export the displayed shortlist. In this way, the dashboard supports the inspection, comparison, and interpretation of the final pathway results.

5 Results

5.1 Evaluation across datasets

Pipelines A and B were applied to the analysed RNA-seq datasets to test whether the framework generalised beyond the original reference dataset. Overall, the pipelines could be applied across datasets with different gene identifier formats, universe sizes, and differential expression patterns. However, the reliability of the final shortlist depended strongly on the properties of the input data.

The overall dataset-level behaviour is summarised visually in Supplementary Figure 3. Pipeline A produced high-confidence Tier 1 results, meaning stable pathways with GO-WP consensus, for `dataset_0` and `PAR1KO_iVSMC`, while `dataset_3` and `ACTA2KO_iVSMC` required the threshold policy to obtain a stronger Tier 1 output. In contrast, `dataset_1` and `Aneurysm_pVSMC` were limited by missing GO-WP consensus, and `dataset_2` was limited by low pathway coverage. This shows that the framework did not fail in one uniform way; instead, different datasets exposed different limitations of enrichment analysis.

Pipeline B generally produced compact directional outputs, usually with ten final pathways. The exception was `PAR1KO_iVSMC`, where Pipeline B produced only three final pathways. This suggested that Pipeline B was useful as a complementary view, but that it could still be limited by the structure of the available expression data. Overall, the multi-dataset evaluation showed

that the pipelines generalised technically, but that confidence in the biological interpretation had to be judged in relation to dataset quality, coverage, and consensus behaviour.

5.2 Threshold automation and dataset classification

The threshold policy showed that fixed thresholds were not equally suitable for all datasets. The default setting worked well for healthy datasets, but it was too strict for borderline datasets and could not solve cases where the limitation came from the input data itself.

The clearest improvement was observed for the borderline datasets. For **dataset_3**, the default consensus threshold produced only 2 Tier 1 pathways, which was below the minimum required for Tier 1 export. After applying the policy, the consensus threshold was relaxed from $\text{CONS_J} = 0.15$ to $\text{CONS_J} = 0.10$, while the stability threshold remained fixed at $\tau = 0.70$. This increased the final Tier 1 output to 8 pathways. A similar pattern was observed for **ACTA2KO_iVSMC**, where the baseline setting produced 4 Tier 1 pathways, while the recommended threshold setting produced 9 Tier 1 pathways. The threshold sensitivity for selected datasets is shown in Figure 2.

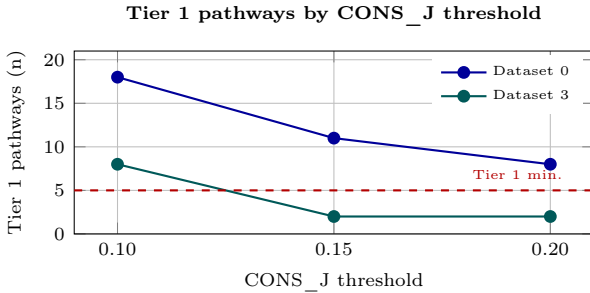


Figure 2: Sensitivity of Tier 1 pathway export to the consensus Jaccard threshold at $\tau = 0.70$. The dashed line marks the minimum of five Tier 1 pathways used for Tier 1 export.

Importantly, the policy did not simply relax thresholds for every dataset. **dataset_1** and **Aneurysm_pVSMC** were classified as no-overlap cases, meaning that stable pathways were present but GO and WP did not share enough genes to form cross-collection consensus. In these cases, relaxing the consensus threshold could not create a reliable Tier 1 result. **dataset_2** was classified as low coverage, because the measured universe and pathway coverage were too limited for reliable enrichment interpretation. This dataset remained a low-confidence case even under more permissive thresholds.

Overall, the threshold policy improved Pipeline A for datasets where the default consensus requirement was only slightly too strict, while also identifying datasets where threshold relaxation would not be meaningful. This shows that dataset-specific threshold selection was useful not only for selecting final results, but also for diagnosing whether the limitation came from threshold choice, missing GO–WP overlap, or insufficient pathway coverage.

The policy outcomes are summarised in Table 1, which gives the final classification and recommended setting for each dataset.

Table 1: Pipeline A threshold policy summary. Base T1 refers to the default threshold setting, while Rec. T1 refers to the policy-recommended setting.

Dataset	Class	Base T1	Rec. T1	Rec. setting
dataset_0	Healthy	11	11	default
dataset_1	No overlap	0	0	no Tier 1
dataset_2	Low coverage	0	0	low conf.
dataset_3	Borderline	2	8	$\text{CONS_J } 0.10$
ACTA2KO	Borderline	4	9	$\text{CONS_J } 0.10$
Aneurysm	No overlap	0	0	Tier 2 only
PAR1KO	Healthy	5	5	default

5.3 Pipeline B ablation

The Pipeline B ablation study showed that different components affected the final candidate pool in different ways. Because the standard Pipeline B output is capped at five pathways per direction, the main analysis used uncapped outputs to see how the full candidate pool changed before the final presentation cap was applied.

The full uncapped Pipeline B produced 55 candidate pathways before applying the top-N cap. Removing GO semantic collapse increased the pool to 63 pathways, but it retained almost all of the full-pipeline pathways. This suggests that semantic collapse had a measurable, but relatively small effect on the uncapped output. In contrast, removing Jaccard overlap clustering changed the output much more strongly. The no-overlap variant retained only 25 pathways and shared very few IDs with the full uncapped pipeline. This indicates that overlap clustering was not only removing redundancy, but also strongly shaping which pathways were selected as final representatives.

The CAMERA-only diagnostic variant produced the largest pool, with 68 pathways, and retained most of the full-pipeline pathways. This shows that CAMERA alone recovered much of the same signal in **dataset_0**. However, this variant was only used as a diagnostic test and not as a replacement for the full pipeline, because the normal Pipeline B selection also uses fgsea agreement and enrichment direction.

The main uncapped ablation results are summarised in Table 2, with a visual version provided in Supplementary Figures 4 and 5.

Table 2: Uncapped Pipeline B ablation summary for **dataset_0**. Jaccard and reference ID retention are calculated relative to the full uncapped Pipeline B output.

Variant	Final pool	Jaccard	Ref. IDs kept
Full	55	1.00	100.0%
No semantic	63	0.84	98.2%
No overlap	25	0.07	9.1%
CAMERA-only	68	0.73	94.5%

Overall, the ablation showed that Jaccard overlap clus-

tering was the most influential component for the composition of the Pipeline B candidate pool. Semantic collapse had a smaller effect, while the final top-N cap mainly acted as a presentation choice rather than a quality filter. These results also show why uncapped analysis was important: when only the final capped shortlist is inspected, some component effects are hidden.

5.4 Pipeline A vs Pipeline B comparison

The comparison between Pipeline A and Pipeline B showed that the two pipelines were complementary rather than redundant. Exact pathway ID overlap was low across the comparison datasets, ranging from 0 to 2 shared pathways. This was expected because the pipelines use different statistical approaches and output structures. Pipeline A depends on a DE gene list and produces stability-based but undirected pathway shortlists, while Pipeline B uses ranked enrichment methods and separates UP and DOWN results.

The main comparison results are summarised in Table 3. A visual summary of the same comparison is provided in Supplementary Figure 6. Two datasets showed complementary behaviour. In `dataset_0`, both pipelines produced interpretable pathway shortlists with some overlap, but Pipeline B added directionality to part of the signal. In `ACTA2K0_iVSMC`, Pipeline A produced a stable Tier 1 shortlist after the threshold policy was applied, while Pipeline B separated the results into UP and DOWN pathway groups. This suggested that Pipeline A was useful for identifying stable themes, while Pipeline B added information about the direction of pathway-level change.

The clearest case where Pipeline B was more informative was `Aneurysm_pVSMC`. This dataset had only 16 DE genes, which limited the ORA-based Pipeline A output to three Tier 2 pathways. Pipeline B still produced a directional shortlist of ten pathways. This does not prove that all Pipeline B pathways are biologically correct, especially because the dataset had only three samples per group. However, it shows that Pipeline B can provide useful hypothesis-generating output in a low-DE setting where ORA has limited power.

The clearest case where Pipeline A was more informative was `PAR1K0_iVSMC`. Pipeline A produced five Tier 1 pathways, while Pipeline B produced only three UP pathways and no exact overlap with Pipeline A. This difference was likely influenced by the input available to each pipeline in this case. Pipeline A used the final differential expression statistics, while Pipeline B used the expression subset available for the contractile comparison. Therefore, the Pipeline B result was more limited for this dataset.

Table 3: Pipeline A and Pipeline B comparison summary. “Comp.” indicates complementary behaviour between the two pipelines.

Dataset	A final	B final	Overlap	Outcome
<code>dataset_0</code>	11	10	2	Comp.
<code>ACTA2K0</code>	9	10	1	Comp.
<code>Aneurysm</code>	3	10	1	B > A
<code>PAR1K0</code>	5	3	0	A > B

Overall, no single pipeline consistently outperformed the other. Pipeline B was most useful when the DE gene list was very small, while Pipeline A was stronger when the upstream DE analysis produced a reliable and well-supported gene list. The shared pathway IDs were rare, but when overlap occurred it provided additional confidence because the finding was supported by two different pipeline designs. This supports the use of both pipelines as complementary views of the same pathway analysis problem.

5.5 Dashboard outcome

The final output of the project was an interactive dashboard for inspecting the pathway shortlists. The dashboard was designed to make the pipeline results easier to explore without manually opening separate output files. It loads precomputed results from Pipeline A, Pipeline B, the threshold policy, and the cross-pipeline comparison. The dashboard can be viewed and run through the accompanying GitHub repository: [GitHub repository](#). Only the publicly available datasets are included in the repository; the full set of project datasets is not shared because some datasets were provided within the project collaboration. Example dashboard views are shown in Appendix B.

The dashboard supports three main tasks. First, it makes inspection easier by showing the final shortlist in a searchable table and by displaying additional information for a selected pathway. Second, it supports interpretation by showing visual summaries such as a dot plot, selection funnel, and overlap network. These views make it easier to see which pathways were selected, how the list was reduced, and whether terms overlap with each other. Third, it supports comparison by allowing the user to switch between datasets and pipelines, and by including an A–B comparison view for datasets where both pipelines were analysed.

Overall, the dashboard made the final results more accessible and easier to communicate. It does not replace the pipeline scripts or rerun the analysis, but it provides a practical layer for exploring the final outputs. This directly addresses the goal of making the pathway results easier to inspect, compare, and interpret.

6 Discussion

The second-semester results show that the pathway analysis framework can be applied across multiple RNA-

seq datasets, but that the reliability of the final shortlist depends strongly on the input data. The pipelines did not fail in one uniform way. Instead, different datasets exposed different limitations, such as low pathway coverage, weak differential expression signal, missing GO–WP consensus, or limited sample access for Pipeline B. This supports the idea that pathway enrichment results should be interpreted together with dataset-level diagnostics, rather than only by looking at the final pathway list.

The dataset-aware threshold policy improved Pipeline A in cases where the default consensus threshold was slightly too strict. This was useful for borderline datasets, because relaxing `CONS_J` while keeping the stability threshold strict recovered stronger Tier 1 outputs. However, the policy also showed that threshold relaxation cannot solve every problem. When the issue was low coverage or missing cross-collection overlap, the output had to remain lower confidence. This is important because it prevents the pipeline from forcing high-confidence results when the data do not support them.

The Pipeline B ablation showed that the final shortlist is strongly affected by redundancy-reduction choices. In particular, Jaccard overlap clustering had a larger effect than expected, because it did not only remove similar pathways but also shaped which terms became final representatives. This means that redundancy reduction is not just a cosmetic step after enrichment; it can influence the biological interpretation of the final shortlist. The uncapped ablation also showed that the top-N cap should mainly be treated as a presentation choice, because it can hide differences between pipeline variants.

The comparison between Pipeline A and Pipeline B showed that the two pipelines are complementary rather than redundant. Pipeline A is stronger when a reliable DE gene list and a good upstream statistical model are available. Pipeline B is more useful when few genes pass the DE threshold, because it can still detect coordinated pathway-level changes in ranked data and provides UP/DOWN direction. However, the biological interpretation of the selected pathways remains a limitation. The project mainly evaluated computational reliability and interpretability, not experimental validation. Therefore, the final pathways should be treated as candidate biological findings that need further expert review.

7 Future Work

Although this project extended the original pathway analysis framework, several directions could still be explored in future work. The most important next step would be to test the framework on more independent RNA-seq datasets and combine this with more detailed expert review of the selected pathways. In this project, the results were discussed with supervisors and generally appeared to make sense, but a larger number of datasets and more biological input would be needed to confirm whether the pipelines work as well as expected

across different contexts.

Another useful direction would be to improve the handling of low-coverage and no-overlap cases. The results showed that some datasets cannot be improved simply by relaxing thresholds. Future work could define clearer warning rules or alternative strategies for datasets where pathway coverage is too low, or where GO and WP do not form enough direct consensus pairs.

The GO–WP consensus logic could also be refined. At the moment, consensus is mainly based on gene-set overlap using the Jaccard index. This is useful, but it may miss cases where GO and WP describe related biological processes using different gene sets. Future versions could combine gene overlap with semantic similarity or other redundancy measures.

Finally, the dashboard could be extended further. The current version supports inspection and comparison of precomputed results, but future work could add more advanced cross-pipeline comparison views, clearer confidence labels, and automatic report export. This would make the framework easier to use as a decision-support tool for interpreting pathway enrichment results.

8 Conclusion

This semester extended the pathway analysis framework from a single-dataset proof of concept to a broader multi-dataset evaluation. The results showed that the pipelines could be applied across different RNA-seq datasets, but that the confidence of the final pathway shortlist depended strongly on dataset properties such as differential expression signal, pathway coverage, and GO–WP consensus.

The dataset-aware threshold policy improved Pipeline A by identifying when threshold adjustment was useful and when the input data itself limited the result. The Pipeline B ablation showed that redundancy-reduction steps, especially overlap clustering, can strongly affect the final candidate pool. The comparison between Pipeline A and Pipeline B showed that the pipelines are complementary: Pipeline A is useful for stable consensus-based pathway themes, while Pipeline B adds directionality and can be more informative when the DE gene list is small.

Overall, the project shows that pathway enrichment results become more interpretable when they are supported by redundancy reduction, stability checks, and dataset-aware diagnostics. The final dashboard provides a practical way to inspect and compare these outputs, making the framework easier to use for exploring RNA-seq pathway results.

9 Self-Reflection

This semester felt more concrete than the first semester because I was able to test the pipelines on new datasets instead of only developing them on the original one. Receiving the blinded study datasets, and later the

datasets from the Biochemistry department, was an important step because it helped test whether the framework could generalise. The results from these datasets gave me more confidence that the project was not only technically working, but also producing outputs that could be interpreted in a meaningful way.

I also improved in presenting and discussing my work. Based on the feedback from the first semester, I tried to share more during meetings and explain my progress more clearly. This was easier because the project became more concrete and I had more results to discuss. Making the poster and presenting it at my first conference was also a valuable experience, because it helped me practise explaining my work outside of the normal project setting.

The hardest part was that this was my first large individual research project. As a Data Science and Artificial Intelligence student, I could work on the pipeline design, evaluation, and visualisation, but the biological context was still challenging and required discussion with people who had more domain knowledge. The technical and research-management parts were also demanding, because there were many possible directions and it was not always obvious which ones were the most useful. Because of this, I had to keep deciding what was realistic within the available time and which results were most important to include.

Compared to the first semester, I became more systematic in comparing results and thinking about how to communicate them. I also became more comfortable with the uncertainty that comes with research, especially when working for some time without knowing whether the method will generalise to new data. One thing I would still improve is my documentation process. This semester included many experiments, outputs, and decisions, and it would have been easier to write the report if I had kept a clearer running record from the beginning.

During this semester, I also used AI tools as support for parts of the project. This was especially helpful for the dashboard, both for structuring the implementation and for thinking through the design of the interface, such as how to organise the outputs, figures, and comparison views in a clearer way. I also used AI support when debugging code, improving figure outputs, and polishing some written explanations. I did not use these tools as a replacement for doing the work myself, but as support while developing, checking, and improving my own implementation. This made the project more manageable and helped me turn the results into something that was easier to inspect and communicate.

Overall, this project gave me my first real taste of research. It was challenging, but also a very good experience. It confirmed that I am interested in bioinformatics and in projects where technical methods are used to support biological interpretation. Going forward, I would like to keep improving both my technical skills and my scientific understanding in this area.

A Supplementary Figures

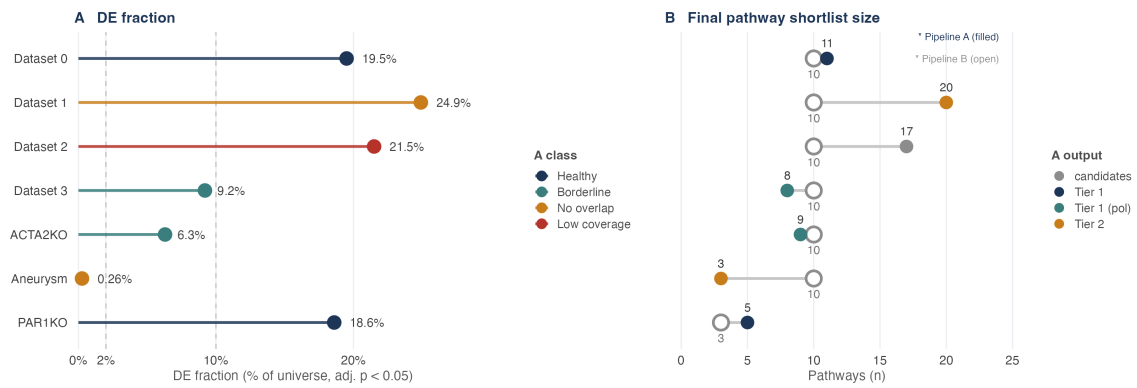


Figure 3: Overview of the analysed datasets, showing DE fraction, Pipeline A classification, and final shortlist sizes for Pipeline A and Pipeline B. This figure supports the multi-dataset evaluation by showing that the datasets differed in signal strength and final pipeline behaviour.

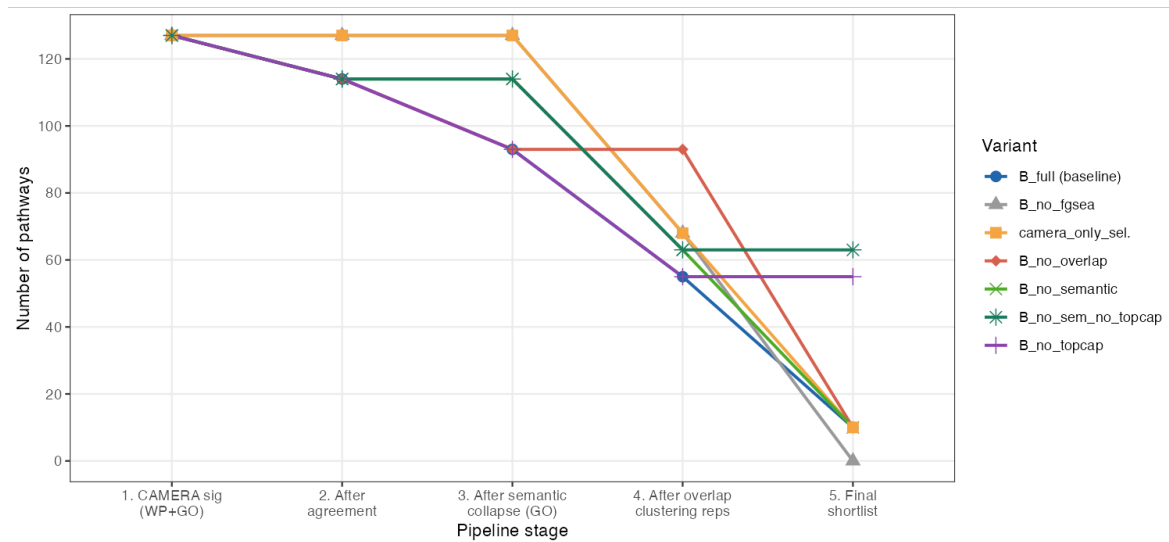


Figure 4: Pipeline B pathway funnel for the uncapped ablation analysis on **dataset_0**. The line graph shows how the number of retained pathways changes across the main Pipeline B stages for the full pipeline and each ablation variant.

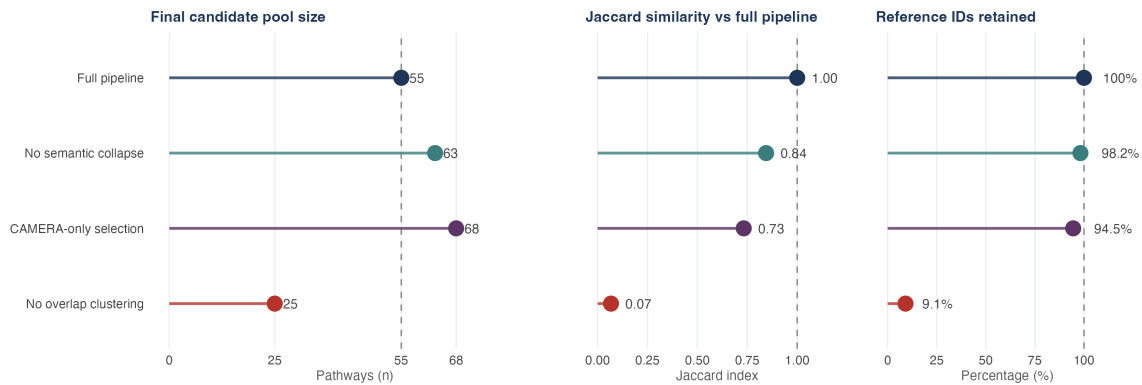


Figure 5: Visual summary of the uncapped Pipeline B ablation on **dataset_0**. The panels compare final candidate pool size, Jaccard similarity to the full pipeline, and the percentage of full-pipeline pathway IDs retained by each ablation variant.

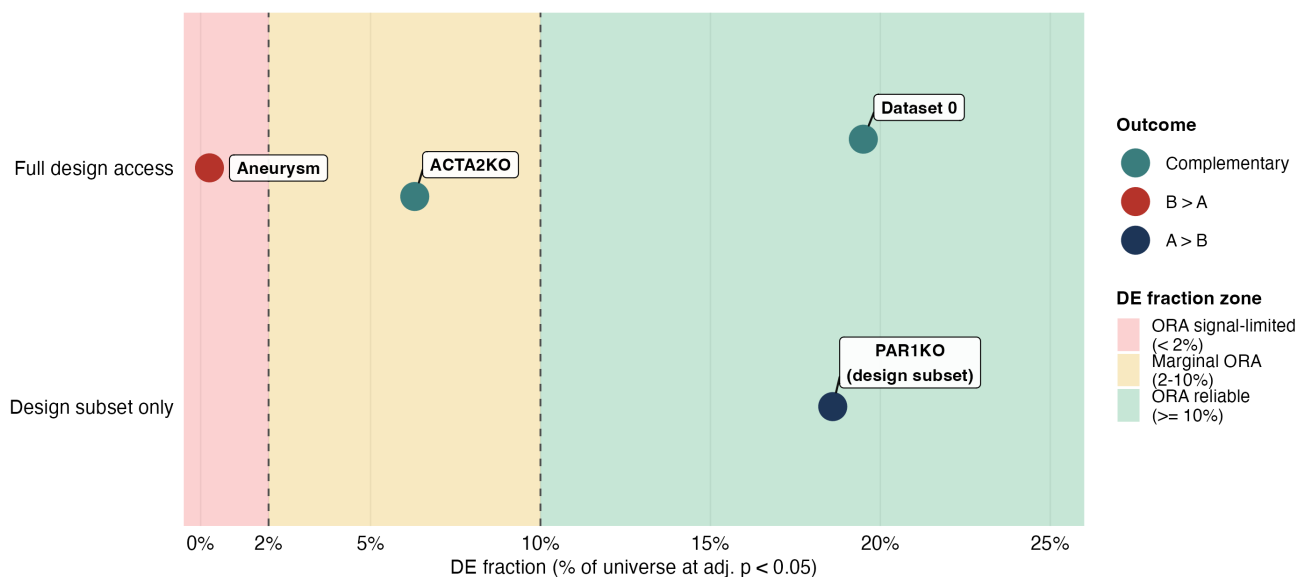


Figure 6: Outcome landscape for the Pipeline A and Pipeline B comparison. Datasets are positioned by DE fraction and by the amount of expression data available for the Pipeline B comparison. Colours indicate whether the pipelines were complementary or whether one pipeline was more informative.

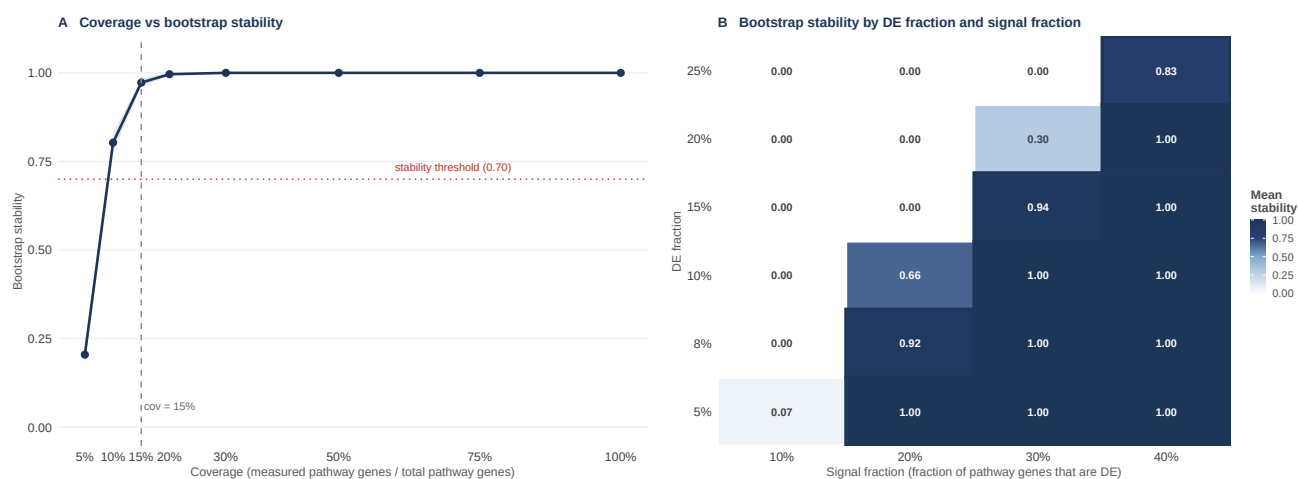
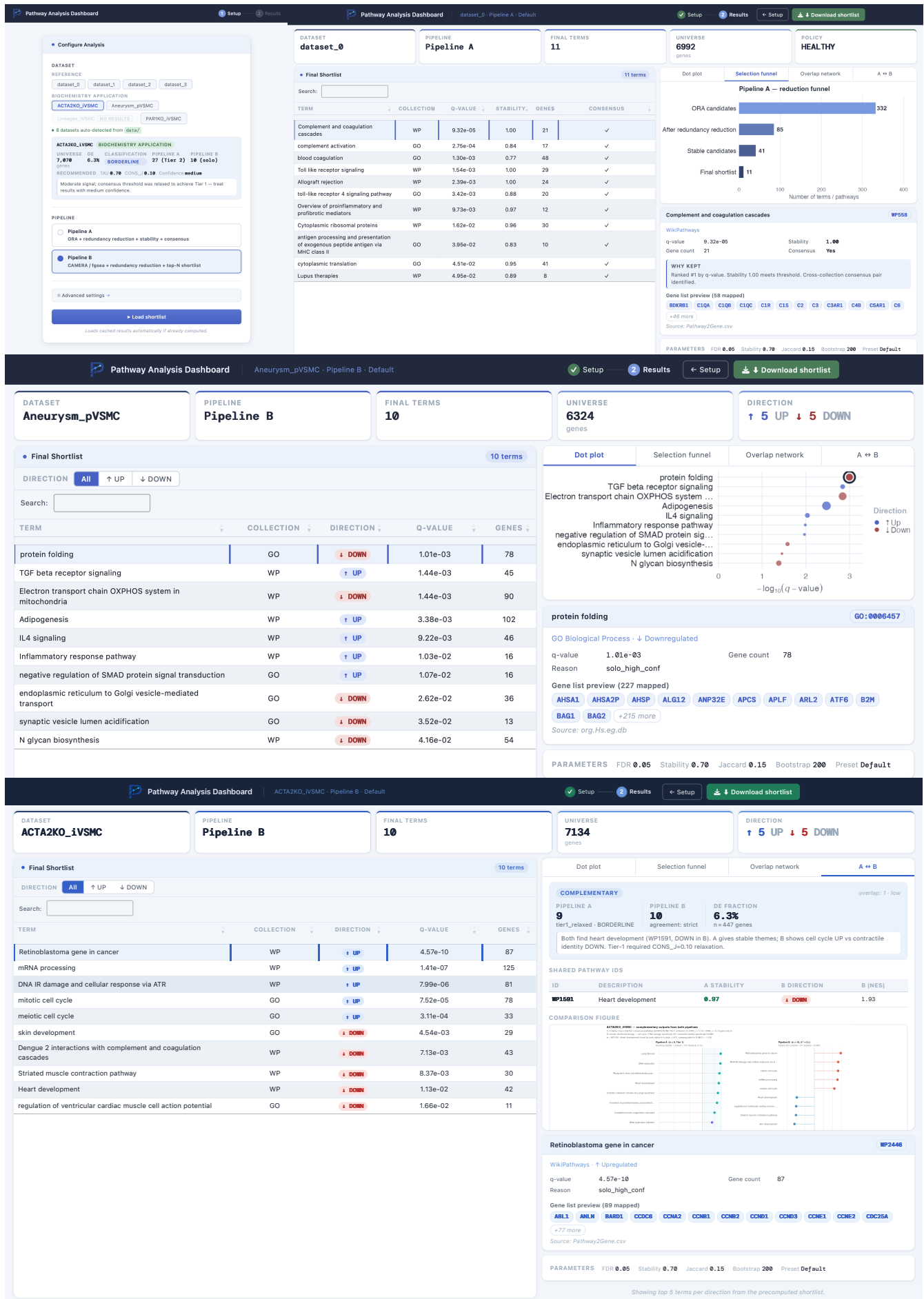


Figure 7: Simulation results used to check threshold behaviour. Panel A shows how bootstrap stability changes with pathway coverage, while Panel B shows mean bootstrap stability across combinations of DE fraction and pathway signal fraction.

B Dashboard



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